

Jesus Morales Villar

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Port Orchard, WA

<https://vilmora-dev.github.io/vilmora-dev/>

PROFESSIONAL SUMMARY

B.S. in Software Engineering with over 3 years building web applications and 1 year developing predictive models, plus hands-on experience in full-stack web development, geospatial data processing, and ML. Skilled in building end-to-end solutions—from data validation and feature engineering to backend integration, React frontends, and production ML pipelines. Fluent in English and Spanish, with a collaborative communication style focused on measurable impact, reproducibility, and scalable, user-focused software solutions.

EDUCATION

B.S. in Software Development Engineer, UnADM, Mexico

(July 2020 - December 2024)

Google IT Support Professional Certificate

(July 2023 - December 2023)

SKILLS

Languages: Fluent English & Spanish

Technical Skills: Full-stack engineering across JavaScript (React, TypeScript, Node.js, Express) and Python (NumPy, Pandas, GeoPandas, scikit-learn, FastAPI), with experience in PostgreSQL, MongoDB, Docker, and Git.

Documentation: Software specification documents, technical help articles, user guides, bilingual content.

Soft Skills: Technical communication, cross-functional collaboration, critical thinking and problem-solving, clear written and verbal communication

PROFESSIONAL EXPERIENCE

Application Developer - Delta Rising Foundation

(August 2025 - May 2026)

- **Grant Discovery Platform** (March 2025 - May 2026) — Laravel 13 + React 18 dashboard backed by a Python scraper across 6 sources (Grants.gov, FEMA HMA, CA Grants Portal, Simpler.Grants.gov, Terra Viva RSS, DuckDuckGo). Applied all-MiniLM-L6-v2 semantic scoring for mission-relevance filtering. Built role-based access, an immutable audit log, and a priority-tiered keyword budget with round-robin initiative rotation keeping daily GitHub Actions runs under 25 minutes.
- **Forest Carbon Modeling Platform** (August 2025 - March 2026) — ML pipeline combining LiDAR, NAIP imagery, and soil/climate data to estimate per-tree carbon sequestration. Trained LightGBM/XGBoost/Random Forest models with 10+ spectral indices (NDVI, EVI, VARI, NDWI, NBR). Trained YOLOv8 for automated canopy detection (72% mAP) and built PDAL/GeoPandas pipelines for reproducible model training across new forest sites.

Full-Stack Developer - COBACH

(April 2024 - October 2024)

Built a centralized project management system to streamline task tracking, deadlines, status updates, and custom support ticketing across COBACH educational programs, improving data accessibility and team collaboration.

- Admin dashboard with diagrams visualizing project progress across departments/teams.
- Kanban board to display all project activities with drag-and-drop functionality.
- Document management system for attaching files and tracking reports per activity.
- Ticketing system for troubleshooting and admin-only actions.

PROJECTS

Explore my work:

10+ projects on <https://github.com/vilmora-dev/> — React, Laravel, PHP, FastAPI, LightGBM, geospatial AI